

Maciej Krzysztof Zuziak

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PROFILE

AI researcher specialising in the security and privacy of decentralised machine learning – federated learning, membership inference attacks, differential privacy, and data leakage. Marie Skłodowska-Curie Doctoral Fellow (under 10% acceptance rate) and Max Weber Fellow (under 3% acceptance rate) with 5+ years of experience delivering EU-funded research (€20M+ programmes) across academia and industry. Combines rigorous ML engineering with formal legal training in data rights and compliance – able both to design privacy-preserving systems and to assess their regulatory implications. Experienced in independent research leadership, interdisciplinary collaboration, and translating theory into deployable decentralised systems.

EXPERIENCE

- European University Institute (EUI)** Sep 2026 – Sep 2027
Max Weber Fellow (Incoming), Department of Law and Robert Schuman Centre for Advanced Studies Florence, Italy
 - Research fellowship awarded for independent research on emerging AI safety guidelines and liability rules governing modern AI systems.
 - Selected as one of 56 distinguished researchers (out of more than 2000 applications) to conduct independent research under the supervision of senior researchers from the EUI Department of Law and Robert Schuman Centre for Advanced Studies.
- University of Leeds** Jan 2026 – Jun 2026
Research Fellow, School of Computer Science Leeds, United Kingdom
 - Led independent research on decentralised systems for Smart Grids, UAV coordination, and Digital Democracy, identifying new methodological directions for the lab.
 - Corresponding author on *The Optimization Trilemma* (arXiv:2607.17311), extending decentralised multi-agent coordination to optimise system efficiency, individual comfort and fairness jointly.
 - Surveyed emerging research areas and co-authored joint grant proposals targeting Horizon Europe, UKRI, and ARIA funding schemes.
 - Mentored junior researchers and coordinated lab-critical projects, contributing to research continuity and delivery.
- National Research Council of Italy (CNR)** Nov 2021 – Dec 2025
Research Associate, Knowledge Discovery Lab Pisa, Italy
 - Core contributor to two Horizon-scale EU projects (€11M+), focusing on federated personalisation, explainability, and privacy risk mitigation.
 - Published 5 peer-reviewed outputs and delivered 1 journal article currently under review.
 - Designed and implemented reusable ML libraries for decentralised learning, including controlled non-IID splits, clustered personalisation, and Shapley-based contribution analysis.
 - Contributed methodological components adopted across multiple project work packages and experimental pipelines.
 - Collaborated with leading European research institutions CNRS, Scuola Normale Superiore, Scuola Superiore Sant'Anna, and other European partners on tractable federated algorithms and data governance frameworks.
- Tellu AS** Sep 2023 – Sep 2024 (dual appointment)
Visiting Research Engineer, DevOps Department Oslo, Norway
 - Co-developed a production-grade last-mile IoT gateway with autonomous onboarding and decentralised fleet and device-health management.
 - Designed decentralised Digital Twin architectures for device enrolment, metrics collection, and lifecycle monitoring.
 - Delivered technical documentation for three Horizon Europe mid-term reviews, all positively evaluated.
- Permanent Mission of the Republic of Poland to the United Nations** Jul 2019 – Aug 2019
Intern, Security Council New York, USA
 - Attended Security Council sessions on peacekeeping, counter-terrorism, and transnational crime; engaged with Sanctions Committees (DR Congo, Libya) and commended by the Head of Mission.

EDUCATION

• University of Pisa

Ph.D. (Artificial Intelligence)

Nov 2021 – Dec 2025

Pisa, Italy

- Thesis: *Data Collaboration with the Use of Federated Learning: Issues of Marginal Contribution Quantification and Personalisation* (evaluation: eccellente - highest distinction). - [Thesis Repository](#).
- Joint National Doctorate in AI coordinated with, *among others*, Scuola Superiore Sant'Anna, Scuola Normale Superiore, University of Trento, and University of Bologna.
- 300+ hours of advanced CS & mathematics modules (graded A–A+).
- All three annual appraisals rated exceptional.

• University of Silesia in Katowice

Master of Laws (LL.M., integrated 5-year programme)

Sep 2016 – Jul 2021

Katowice, Poland

- Thesis: *Digitalisation of Factoring Contracts* (evaluation: bardzo dobry - highest distinction).
- Post-graduation prize for most promising graduates.

JOURNALS

- [1] [\[First\]](#) M.K. Zuziak, R. Pellungrini, S. Rinzivillo (2025): **One-Shot Clustering for Federated Learning Under Clustering-Agnostic Assumption**. ACM Transactions on Intelligent Systems and Technology (revision under review) - [ArXiv Preprint](#).

PUBLICATIONS

FIRST = FIRST AUTHOR; CO-FIRST = EQUAL CONTRIBUTION, AS MARKED IN THE PUBLICATION

- [1] [\[Co-first\]](#) S.Z. El Mestari*, M.K. Zuziak*, G. Lenzini (2026): **Poison to Detect: Detection of Targeted Overfitting in Federated Learning**. *Proceedings on Privacy Enhancing Technologies* 2026(4), pp. 480–506. Presented at *The 26th Privacy Enhancing Technologies Symposium*, Calgary, Canada - [PoPETs Access](#). (*equal contribution)
- [2] [\[Co-first\]](#) S.Z. El Mestari*, M.K. Zuziak*, G. Lenzini, S. Rinzivillo (2025): **Can Contributing More Put You at a Higher Leakage Risk? The Relationship Between Shapley Value and Training Data Leakage Risks in Federated Learning**. At *SECRYPT 2025*, Bilbao, Spain. SciTePress - [SciTePress Access](#). (*equal contribution)
- [3] [\[First\]](#) M.K. Zuziak, R. Pellungrini, S. Rinzivillo (2024): **One-Shot Clustering for Federated Learning**. At *IEEE Big Data 2024*, Washington, USA. IEEE Xplore - [IEEE Xplore Access](#).
- [4] [\[First\]](#) M.K. Zuziak, S. Rinzivillo (2024): **Amplified Contribution Analysis for Federated Learning**. At *IDA 2024*, Stockholm, Sweden. Springer Lecture Notes in Computer Science (LNCS) - [Springer Nature Access](#).
- [5] [\[Co-first\]](#) M.K. Zuziak*, O. Hinrichs*, A. Abdrassulova, S. Rinzivillo (2023): **Data Collaboratives with the Use of Decentralised Learning**. At *ACM FAccT 2023*, Chicago, USA. ACM - [ACM Digital Library Access](#). (*equal contribution)

PREPRINTS

- [1] J. Nikolic, M.K. Zuziak, E. Pournaras (2026): **The Optimization Trilemma: Efficiency, Comfort and Fairness in Decentralized Multi-agent Coordination**. arXiv:2607.17311 - [ArXiv Preprint](#).

POLICY CONTRIBUTIONS

- [1] ACM ETPC (2026). **Comments in Response to European Commission Call for Evidence Survey on the “Digital Omnibus on AI Regulation” Proposal**. Contributors: G. Besanson, M.K. Zuziak, A. Saucedo, F. Medeiros, T. Romanoff. [ACM ETPC Link](#).
- [2] ACM ETPC (2025). **Consultation on the Guidelines on Protection of Minors Online under the Digital Services Act**. Contributors: T. Romanoff, A. Saucedo, M.K. Zuziak, M. Beaudouin-Lafon, C.L. Hankin. [ACM ETPC Link](#).
- [3] ACM ETPC (2024). **Comments on Draft Delegated Regulation on Data Access (DSA)**. Contributors: A. Deshpande, L. DiBiasi, M.K. Zuziak, J. Rozanec, A. Nagy, A. Saucedo, F. Medeiros, J. Mizeria-Petraszko. [ACM ETPC Link](#).

WORKSHOPS, POSTERS & OPINIONS

W=WORKSHOPS, P=POSTERS, O=OPINIONS

- [P.3] S.Z. El Mestari*, M.K. Zuziak*, G. Lenzini (2026): **Poison to Detect: Detection of Targeted Overfitting in Federated Learning (Poster Version)**. Privacy Enhancing Technologies Symposium, Canada. (*equal contribution).
- [P.2] M.K. Zuziak, S. Rinzivillo (2024): **Personalization via Clustering in Federated Learning for Accessible Digital Services**. School of Advanced Studies Sant'Anna at Pisa Internet Festival, Italy.
- [O.1] M.K. Zuziak (2024): **How to Collaboratively Use Statistical Models in a Secure Way**. *Opinio Juris in Comparison*. [Opinio Juris Link](#).

- [P.1] M.K. Zuziak, S. Rinzivillo (2022): **Federated Learning for Personal Data Management under European Digital Strategy**. Conference on Data Sciences and EU Regulations, IRIT at Université Toulouse III, France.
- [W.1] M.K. Zuziak, S. Rinzivillo (2022): **Federated Learning as an Analytical Framework for Personal Data Management - a Proposition Paper**. Hybrid Human Artificial Intelligence (HHAI) Workshop, Amsterdam, Netherlands. CEUR Proceedings - [CEUR WS Link](#).

RECENT TALKS AND SEMINARS

- [1] Institute of Science Tokyo (2026): **On the Intrinsic Vulnerabilities in Federated Learning and Corresponding Defence Methods**.
- [2] ACM (2026): **Europe's Digital Omnibus: A New Digital Regime or Simplification**. [ACM Recording](#).

SELECTED PROJECTS

- **Project Tango (Horizon Europe No. 101120763)** 2024 – 2025
Research Associate  [Project page](#)
 - **Total Funding:** €7M
 - **Key Partners:** Université Paris Cité, TU Darmstadt, University of Pisa, Scuola Normale Superiore.
 - **Role:** Research on clustered federated learning and explainability for human-centric decision-making.
- **Project LEADS (MSCA ITN No. 956562)** 2021 – 2024
Marie Skłodowska-Curie Actions (MSCA) Doctoral Fellow  [Project page](#)
 - **Total Funding:** €3.8M
 - **Key Partners:** University of Pisa, Scuola Superiore Sant'Anna, CNRS, University of Luxembourg.
 - **Role:** Research on trustworthy and decentralised ML infrastructure for the Digital European Strategy.
- **Project ENTRUST (Horizon Europe No. 101095634)** 2023 – 2024
Visiting Contributing Researcher  [Project page](#)
 - **Total Funding:** €5.6M
 - **Key Partners:** Tellu AS, TU Eindhoven, Siemens, Sintef.
 - **Role:** Contributed to decentralised trust management architecture; designed risk assessment processes for critical IoT infrastructure (last-mile medical gateways).
- **Project Eratosthenes (Horizon 2020 No. 101020416)** 2023 – 2024
Visiting Contributing Researcher  [Project page](#)
 - **Total Funding:** €6M
 - **Key Partners:** Tellu AS, TU Graz, KU Leuven, Airbus Cybersecurity, Sintef.
 - **Role:** Contributed to an autonomous onboarding flow for critical IoT infrastructure (last-mile medical gateways and sensors connected to them).

OPEN SOURCE

-  **FedJust** 2021–2024
Federated Learning toolkit for rapid prototyping and reproducible evaluation of novel federated algorithms. [GitHub](#)
-  **OCFL Suite** 2024–2025
Experimental framework for designing and benchmarking Federated Clustered Learning methods. [GitHub](#)

AWARDS

- **Max Weber Fellowship** 2026
European University Institute (under 3% acceptance rate)
- **Horizon Europe (Project Tango) Fellowship** 2024
National Research Council of Italy (CNR)
- **Marie Skłodowska-Curie Actions Fellowship** 2021
European Commission (under 10% acceptance rate)
- **CSC Scholarship** 2020
China Scholarship Council
- **Academic Scholarship (Top 5%)** 2017
University of Silesia (awarded multiple times 2017–2020)

MEMBERSHIPS

- **ACM Europe Technology Policy Committee (ETPC)** Dec 2023 – Present
Non-voting member

CERTIFICATIONS

- **NVIDIA Building Transformer-Based NLP Applications** 2025
- **LangChain Building Ambient Agents with LangGraph** 2025
- **LangChain Project: Deep Research with LangGraph** 2025

SUMMER SCHOOLS & ACADEMIC EXCHANGES

- **University of Pisa** 2022
Summer School in Artificial Intelligence Pisa, Italy
- **China University of Political Science and Law** 2020
Academic Exchange Programme Beijing, China

RESEARCH INTERESTS

- Federated and decentralised learning
- Privacy-preserving ML (differential privacy, membership inference, data leakage)
- Explainable AI and contribution analysis in distributed settings
- AI governance, liability, and regulation

SKILLS

- **Core Expertise:** Federated Learning, Privacy-Preserving ML, Explainable AI, Data Governance
- **Programming:** Python, Julia, C/C++, TypeScript
- **ML & Data:** PyTorch, TensorFlow/Keras, JAX, Scikit-Learn, NumPy, Pandas
- **Distributed & Systems:** Flower, PySyft/PyGrid, TensorFlow Federated, Docker, Kubernetes, Linux/WSL
- **Agentic & LLM Systems:** LangChain, LangGraph, RAG pipelines, vector databases (FAISS, Pinecone, Weaviate), LoRA/PEFT

ADDITIONAL INFORMATION

Languages: Polish (Native), English (C2), Chinese (B2), Italian (B1/B2)